# SESSION 2: WORKING WITH VARIABLES AND RESOURCES

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## Structure of .tf Files

**Terraform blocks**

Terraform uses several types of blocks to describe infrastructure as code. Each block has a specific purpose and plays a fundamental role in configuring the cloud environment.

Example of blocks: Provider, Resource, variables, locals, output, module, etc.

**Provider Definitions**

Providers specify the cloud or on-premise platforms managing the infrastructure resources

**Infrastructure Resources**

Resources in .tf files define the infrastructure components to be created and managed.

**Variables and Outputs**

Variables allow customization, while outputs export useful information from the configuration.

mycode.tf

C: > Users > rameh > Downloads > mycode.tf > ...

```
# Provider declaration
provider "azurerm" {
  features {}
}

# Input variables
variable "location" {
  description = "Azure region to deploy resources"
  type        = string
  default     = "East US"
}

locals {
  rg_name = "rg-terraform-demo"
}

variable "storage_accounts" {
  description = "list of Storage Account names"
  type        = list(string)
  default     = ["stterraform01", "stterraform02"]
}

# Resource Group
resource "azurerm_resource_group" "rg" {
  name     = local.rg_name
  location = var.location
}

# Storage Accounts using loop (for_each)
resource "azurerm_storage_account" "storage" {
  for_each = toset(var.storage_accounts)

  name                     = each.value
  resource_group_name      = azurerm_resource_group.rg.name
  location                 = azurerm_resource_group.rg.location
  account_tier             = "Standard"
  account_replication_type = "LRS"

  tags = {
    environment = "demo"
  }
}
```

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# Terraform Settings Block

**terraform { }**

- Defines **global settings** for a Terraform configuration.
- Declares **required providers** and their versions.
- Configures the **backend** for state storage and locking.
- Sets Terraform CLI constraints (minimum required version).
- **Common elements:**
- required_version
- required_providers
- backend

"Which provider and version do I need?

```
terraform {
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 3.0"
    }

    azurecaf = {
      source  = "aztfmod/azurecaf"
      version = "~> 1.2"
    }
  }
  required_version = ">= 1.3.0"

  backend "local" {
    path = "relative/path/to/terraform.tfstate"
  }
}
```

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# Provider Block

**provider { }**

- Configures **how Terraform connects** to a specific platform or API.
- Defines **authentication, subscription/account, and regional settings**.
- Uses provider plugins declared in required_providers.
- Can be defined multiple times using **aliases**.

**Typical provider configuration includes:**

- Credentials and authentication method
- Subscription / account context
- Region or location
- Provider-specific features

**Example of providers:**

Azure, AWS, GCP, etc.

"How do I connect and use that provider?"

```
# Provider declaration
provider "azurerm" {
  features {}
  subscription_id = "subscription_id"
  tenant_id       = "tenant_id"
  client_id       = "principal_id"
  alias           = "azurerm-1"


}
```

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# Terraform Version Constraints

- Specify the minimum and/or maximum Terraform version your configuration supports.
- Ensures **compatibility** with the configuration and provider features.
- Declared inside the terraform {} block
- Helps prevent breaking changes in team environments and CI/CD pipelines.
- Supports operators:

  = : exact version

  >= : minimum version

  <= : maximum version

  ~> : compatible version, e.g.,

  ~> 1.5 → >= 1.5.0 and < 2.0.0

```
terraform {
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 3.0"
    }

    azurecaf = {
      source  = "aztfmod/azurecaf"
      version = "~> 1.2"
    }
  }
  required_version = ">= 1.3.0"
```

**> Terraform version**

```
PS C:\Users\rameh\kind> terraform version
Terraform v1.9.8
on windows_amd64

Your version of Terraform is out of date! The latest version
is 1.14.3. You can update by downloading from https://www.terraform.io/downloads.html
PS C:\Users\rameh\kind>
```

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# Resource block

- The resource block is used to define the resources that will be provisioned in the cloud provider.
- It includes resource-specific attributes, such as name, location, and detailed settings.
- Resource blocks are declared in the following format:

Resource "type" "name" {
  parameter_a = "abcdefgh"
  parameter_b = 123456
}

```
resource "azurerm_virtual_network" "example" {
  name                = "example-network"
  address_space       = ["10.0.0.0/16"]
  location            = azurerm_resource_group.example.location
  resource_group_name = azurerm_resource_group.example.name
}
```

- **Resource**: keyword
- **Type**: type of resource. E.g.: azurerm_resource_group
- **Name:** arbitrary and unique name for internal reference of terraform
- **Parameters:** attributes that might vary from resource to resource. Some are optional and some are mandatory

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# Variable blocks: Input

**Variables {}**

- Serve as parameter for configuration of resources on your Terraform code
- Can have the following elements:
  - **Default values**: Will be used if user does not set any value for the variable
  - **Descriptions**: Brief explanation of the purpose of the variable (best practice)
  - **Type constraints**: specify which type is accepted for the variable

**Primitive Types:**

- string
- Number
- bool

```
# Input variable types
variable "location" {
  description = "Azure region for resource deployment"
  type        = string
  default     = "East US"
}

variable "instance_number" {
  description = "Number of container instances"
  type        = number
  default     = 3
}

variable "boolean_example" {
  description = "Example of boolean var"
  type        = bool
  default     = false
}
```

https://spacelift.io/blog/terraform-map-variable

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# Variable blocks: Input

**Collection Types**

Accept any type of elements, as long as every elements is the same type

- **List**: ordered collection
- **Map**: key-value mapping. Support:
  - Map(string)
  - Map(number)
  - Map(list)
  - Map(object({…}))
  - Etc.

**Complex Types**

Each element can have its own type

- **Tuple:** a sequence of elements with fixed length.
- **Object:** a collection of structured, mixed type elements

```
# Example of list
variable "environment" {
  description = "Environment name"
  type        = list(string)
  default     = ["dev", "prod", "qa"]
}

variable "tuple_example" {
  description = "Sequence of elements with different types"
  type        = tuple([string, number, bool])
  default = ["a", 15, true]
}

# object example
variable "server_config" {
  description = "Environment-specific configuration"
  type = object({
    name          = string
    instance_type = string
    max_size      = number
  })
}

# Example of map of objects
variable "env_config" {
  description = "Environment-specific configuration"
  type = map(object({
    instance_type = string
    min_size      = number
    max_size      = number
  }))

  default = {
    dev = {
      instance_type = "t3.small"
      min_size      = 1
      max_size      = 2
    }
    prod = {
      instance_type = "t3.large"
      min_size      = 3
      max_size      = 6
    }
  }
}
```

# Variable blocks: Locals

**Locals {}**

- Used to define internal computed values within a module.
- Help reduce code duplication and simplify references.
- Can be called on the code by **local.variable_name**

```
# Local variables
locals {
  resource_group_name = "rg-${var.environment}-rg"

  region = "northeurope"

  tags = {
    owner = "Infra team"
    contact = "infra@corp.com"
    environment = var.environment
  }
}
```

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# Variable Interpolation

- Interpolation in Terraform allows you to dynamically insert values from variables, data sources, or function calls into your configuration strings.
- Interpolation is done using the ${...} syntax.
- Ex: ${var.region} inserts the value of the variable named **region**

```
variable "environment" {
  default = "prod"
}

resource "azurerm_resource_group" "rg_prod" {
  name     = "rg-${var.environment}"
  location = "westeurope"
}
```

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# .tfvars files

- **.tfvars** files are used to assign values to input variables.
- Separate configuration data from Terraform code.
- Commonly used to manage environment-specific values (dev, test, prod).
- Terraform automatically loads when named:
  - terraform.tfvars
  - *.auto.tfvars
- Terraform follows this precedence order for variables:

1. CLI -var
2. .tfvars / *.auto.tfvars
3. Default values (variables.tf)

Variables can be set using command line:

**> terraform apply –var="instance_number=5"**

Or with configuration files ***.tfvars** or ***.auto.tfvars**

**> terraform apply -var-file="prod.tfvars"**

terraform.tfvars

C: > terraform > terraform.tfvars > ...

```
location        = "East US"
environment     = "prod"
instance_number = 5
```

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# Output Block

**Output { }**

- output blocks expose information from Terraform configurations or modules.
- Useful for sharing values between modules or displaying important info after apply.

Can define:

- **value** → the actual output
- **description** → optional description
- **sensitive** = true → hides secrets

```
output "db_instance_name" {
  value = aws_db_instance.main.name
  description = "DB Instance Name"
}

output "database_password" {
  value     = aws_db_instance.main.password
  sensitive = true
}
```

```
~/Devops/terraform_test$ terraform apply tfplan
azurerm_resource_group.rg: Creating...
azurerm_resource_group.rg: Still creating... [00m10s elapsed]
azurerm_resource_group.rg: Creation complete after 14s [id=/subscriptions/b0c2c839-bd2d-4f80-bfe9-52b2828de1e4/resourceGrou
azurerm_virtual_network.vnet: Creating...
azurerm_virtual_network.vnet: Creation complete after 6s [id=/subscriptions/b0c2c839-bd2d-4f80-bfe9-52b2828de1e4/resourceGr
rs/Microsoft.Network/virtualNetworks/vnet-demo]
azurerm_subnet.subnet: Creating...
azurerm_subnet.subnet: Creation complete after 9s [id=/subscriptions/b0c2c839-bd2d-4f80-bfe9-52b2828de1e4/resourceGroups/rg
osoft.Network/virtualNetworks/vnet-demo/subnets/subnet-demo]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

resource_group_location = "westeurope"
resource_group_name = "rg-demo"
```

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# Module block

- Modules are **reusable, composable Terraform configurations**.
- Allow encapsulation of resources for better maintainability.
- Can be sourced from:
  - Local paths (./modules/...)
  - Git repositories
  - Terraform Registry

```
module "storage_account" {
  source              = "./modules/storage"
  location            = var.location
  resource_group_name = local.rg_name
}
```

```
Dev
  ├── jenkins-using-packer.tfvars
  ├── main.tf
  └── variable.tf
Module
  ├── Ec2andLoadBalancer
  │   ├── main.tf
  │   └── variable.tf
  ├── ECS
  │   ├── main.tf
  │   ├── output.tf
  │   └── variable.tf
  ├── Networking
  │   ├── main.tf
  │   ├── output.tf
  │   └── variable.tf
  └── SecurityGroup
      ├── main.tf
      ├── output.tf
      └── variable.tf
```

Fonte: https://medium.com/xebia-engineering/best-practices-to-create-organize-terraform-code-for-aws-2f4162525a1a

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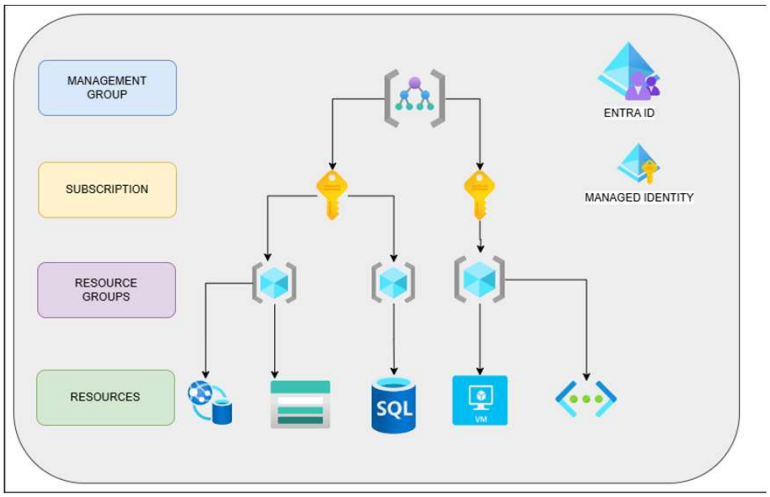
# Azure Overview

- Management Groups
- Subscriptions
- EntraID
- Resource Groups
- Resources
- Service Principals

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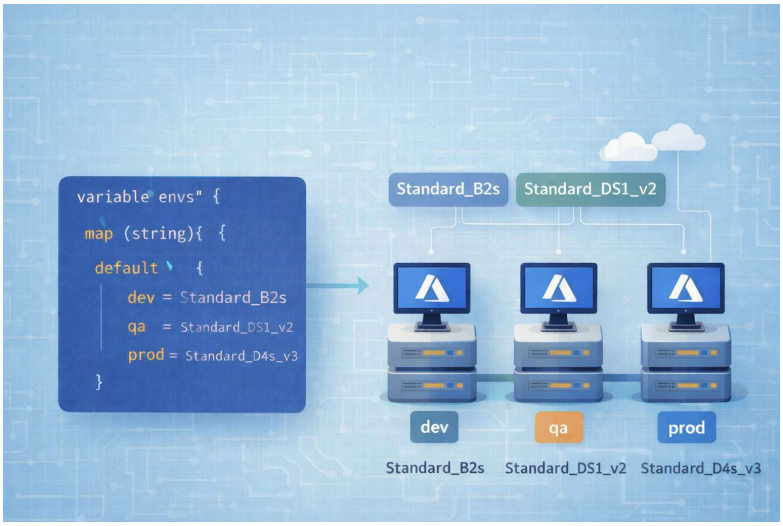
# Lab 2: Building Provisioning Structures Using Variables



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